Sahaj Bhadja

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Summary

Software engineer pursuing an MSCS with 2+ years building C++/Python systems, networked services, automated tests, and real-time operational workflows across Linux-oriented, distributed engineering environments.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Analysis of Algorithms, Database Systems, Web Technologies, Advanced NLP

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Software Engineering, Operating Systems, Object-Oriented Programming, Data Retrieval

Experience

Aasma Technology Solutions **India**
Software Developer Engineer - (SecureSync) *August 2023 – July 2024*
- Engineered **cross-platform integration services** using Python, SQL, AWS Glue, and S3, improving response time by **30%** while supporting distributed APIs, data movement, reporting workflows, debugging, and reliable release coordination across operating environments.
- Maintained **20+ automated validation suites** with Jenkins, PyTest, and Mockito, increasing CI reliability by **68%** and reducing regressions across Windows, Linux, and macOS environments while strengthening readiness, defect isolation, and engineering handoffs.

Aasma Technology Solutions **India**
Full Stack Developer Intern - (VibrantChem) *May 2023 – August 2023*
- Built **REST and WebSocket modules** over SQL/NoSQL persistence, supporting 2M+ synchronized records across integrations, dashboards, and operational workflows while strengthening high-volume reliability, service coordination, data consistency, and issue resolution for active users.
- Automated **Docker-based build and test flows** with Jenkins, cutting release time by **50%** while improving deployment consistency, troubleshooting speed, validation coverage, and team handoffs during recurring releases, debugging cycles, and production-readiness checks.

USC Dornsife **Los Angeles, CA**
Software Engineer *September 2025 – Present*
- Built **Azure SQL-backed backend services** for 500+ users, translating product requirements into APIs, validation logic, schema updates, and production releases across cloud-hosted data workflows, internal tools, stakeholder requests, and maintainable delivery pipelines.
- Improved **service reliability and performance** by tuning indexed queries, background jobs, and CI/CD releases, reducing dashboard latency from **1.2s to 350ms** while supporting operational debugging, release handoffs, traceability, and maintainable incident follow-through.

Grad8 **Los Angeles, CA**
Software Engineer Intern *June 2025 – August 2025*
- Developed **Python services and REST APIs** handling 10,000+ daily requests, integrating uploads, OCR processing, persistence workflows, and automated checks to improve production reliability, validation coverage, and operational stability under active usage.
- Debugged **runtime and workflow defects** across upload, OCR, persistence, and grading failures, documenting fixes and reducing feedback turnaround by **60%** while improving production support, root-cause analysis, and resolution for users and operators.

USC AD Labs **Los Angeles, CA**
ML Research Assistant *January 2025 – May 2025*
- Built **evaluation pipelines** across 100,000+ records, producing structured metrics, logs, and reports for reproducible analysis, anomaly investigation, reliable model validation, and clearer technical reviews across large-scale research and testing workflows.
- Documented **technical workflows and review outputs** for researchers, turning ad hoc experiments into repeatable processes with clearer debugging notes, validation artifacts, traceability, and stronger handoffs across collaborating teams working on iterative evaluations.

Skills

Languages: **C++**, **Python**, Java, JavaScript, TypeScript, SQL, Bash, HTML/CSS
Systems: STL, object-oriented design, Linux/Unix familiarity, IP/networking fundamentals, algorithms, design patterns
Backend: REST APIs, WebSockets, Flask, FastAPI, Node.js, distributed workflows, real-time data handling
Delivery: Docker, Jenkins, Git/GitHub, unit testing, code reviews, CI/CD, Agile/Scrum
Data: PostgreSQL, MySQL, MongoDB, Azure SQL, AWS Glue, S3, troubleshooting, technical documentation

Projects

CPU Scheduling Simulator Live | *C++, React, JavaScript, OS Algorithms*
- Implemented **FCFS, SJF, SRT, RR, priority, and MLQ** schedulers with C++ logic, debugging workflows, validation, timing analysis, and interactive visualization across simulated workloads, strengthening operating-systems reasoning, correctness checks, and performance-focused testing practices.

FinNexus Live | *Next.js, TypeScript, Kubernetes, MongoDB*
- Built a **secure real-time platform** supporting 50K+ users and 1M+ transactions, integrating live updates, tokenized access controls, CI/CD workflows, and visibility dashboards emphasizing responsiveness, reliability, coordinated releases, and secure day-to-day platform operations.

CaptionCraft Live | *Transformers, PyTorch, Flask, Computer Vision*
- Benchmarked **BLIP, GIT, ViT-GPT-2, and custom vision models** on 10K images and 100K captions, improving caption accuracy from **81%** to **96%** through repeatable evaluation pipelines, structured error analysis, and disciplined model-comparison workflows.